



Massachusetts Institute of Technology

Mapping the quantum limits of classical simulatability

L. Shackleton, Phys. Rev. Lett. 136, 186503 (2026)

L. Shackleton, in progress

Emergent phenomena in many-body systems

$$i\hbar \frac{\partial}{\partial t} |\psi\rangle = \hat{H} |\psi\rangle$$



How do we simulate quantum many-body systems?

Simulating quantum systems is **hard**

Quantum Phase Transition at Nonzero Doping in a Random t - J Model

[Leyna Shackleton](#) ¹, [Alexander Wietek](#) ², [Antoine Georges](#) ^{2,3,4,5}, and [Subir Sachdev](#) ¹

The spectral function for the random t - J model, calculated using the Lanczos algorithm on an 18-site cluster, is shown for several values of doping in Fig. 1. A

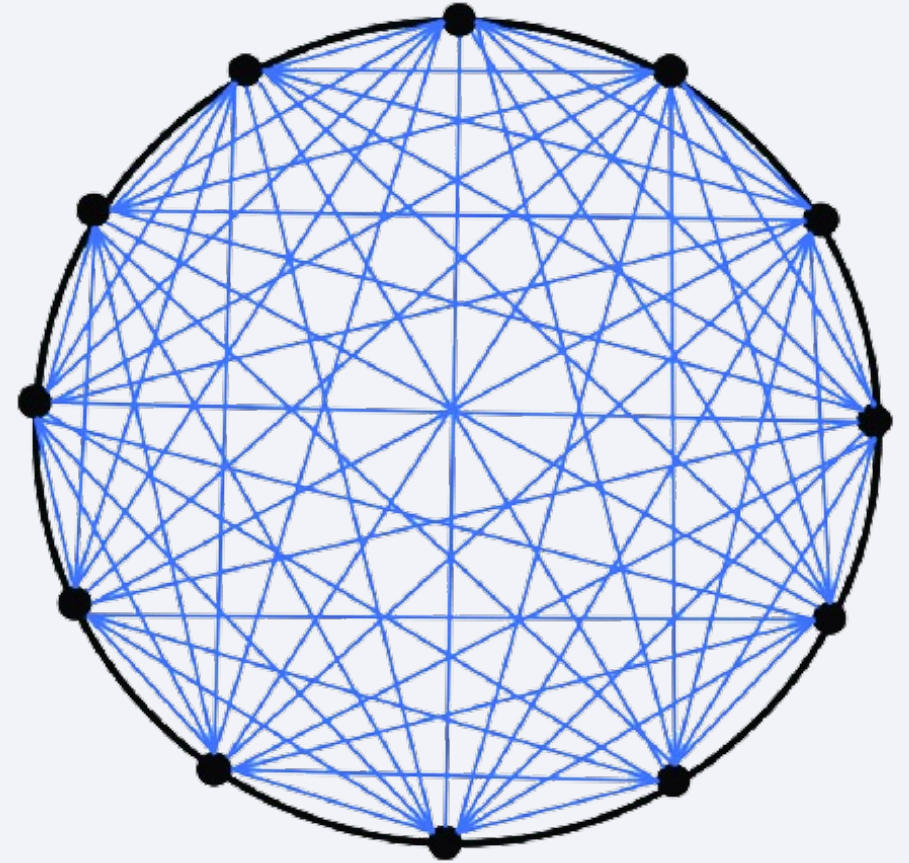


Simulating quantum systems is **hard**

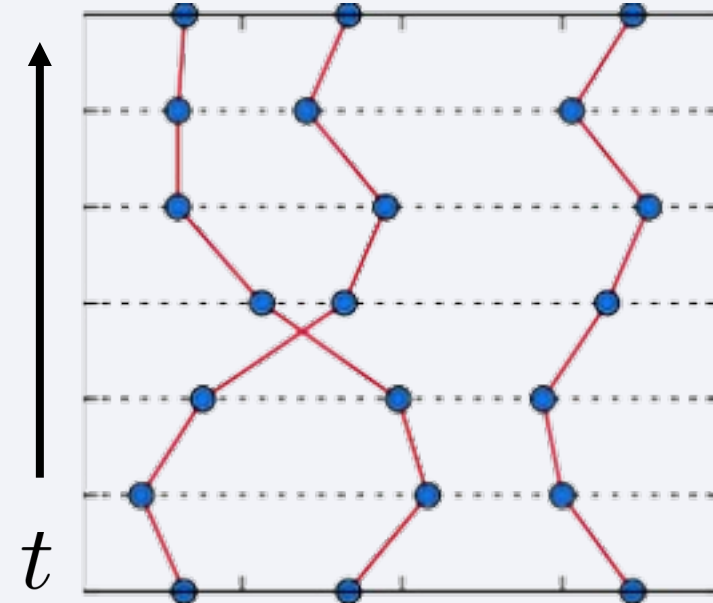
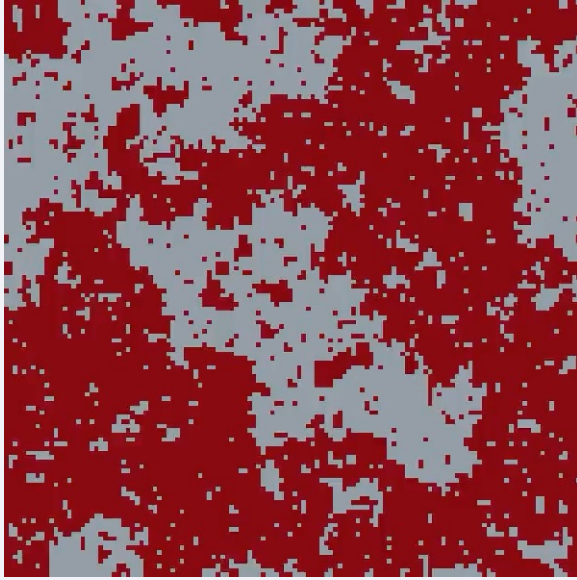
Quantum Phase Transition at Nonzero Doping in a Random t - J Model

[Leyna Shackleton](#) ^{id}¹, [Alexander Wietek](#) ^{id}², [Antoine Georges](#) ^{id}^{2,3,4,5}, and [Subir Sachdev](#) ^{id}¹

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Simulating classical systems is (more) easy



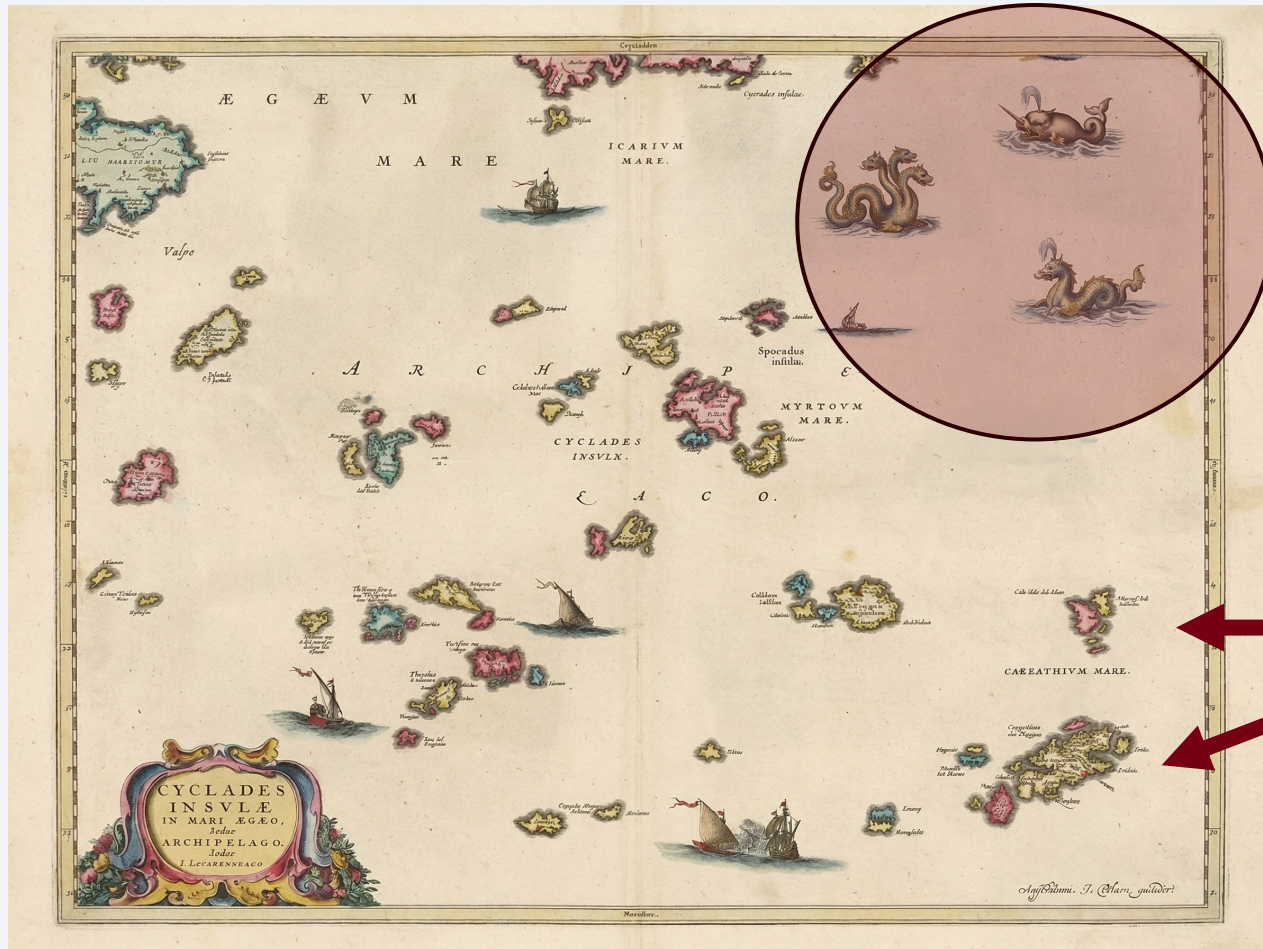
Sample via Monte Carlo

$$Z_{\text{classical}} = \sum_x e^{-E(x)/k_B T}$$

Complex weights

$$Z_{\text{quantum}} = \text{Tr} \left[e^{-\hat{H}/k_B T} \right] \text{ (usually)}$$

The landscape of quantum phases



No known sign problem-free realizations

Sign problem-free islands

Can a phase have an intrinsic sign problem?

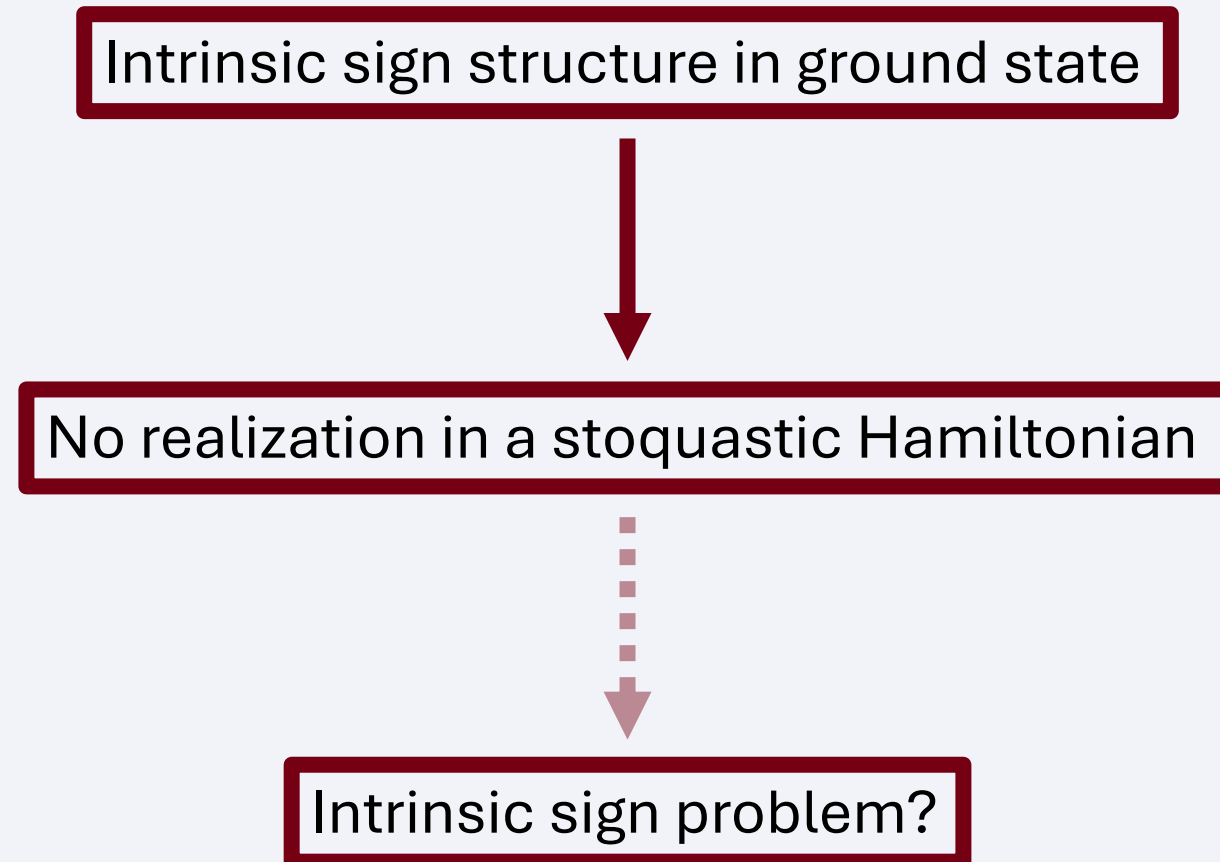
$$Z_{\text{quantum}} = \text{Tr} \left[e^{-\hat{H}/k_B T} \right]$$



Completely positive matrix: “stoquastic”

Stoquastic Hamiltonians must have a completely positive ground state (Perron-Frobenius theorem)

Ground state non-positivity as a proxy

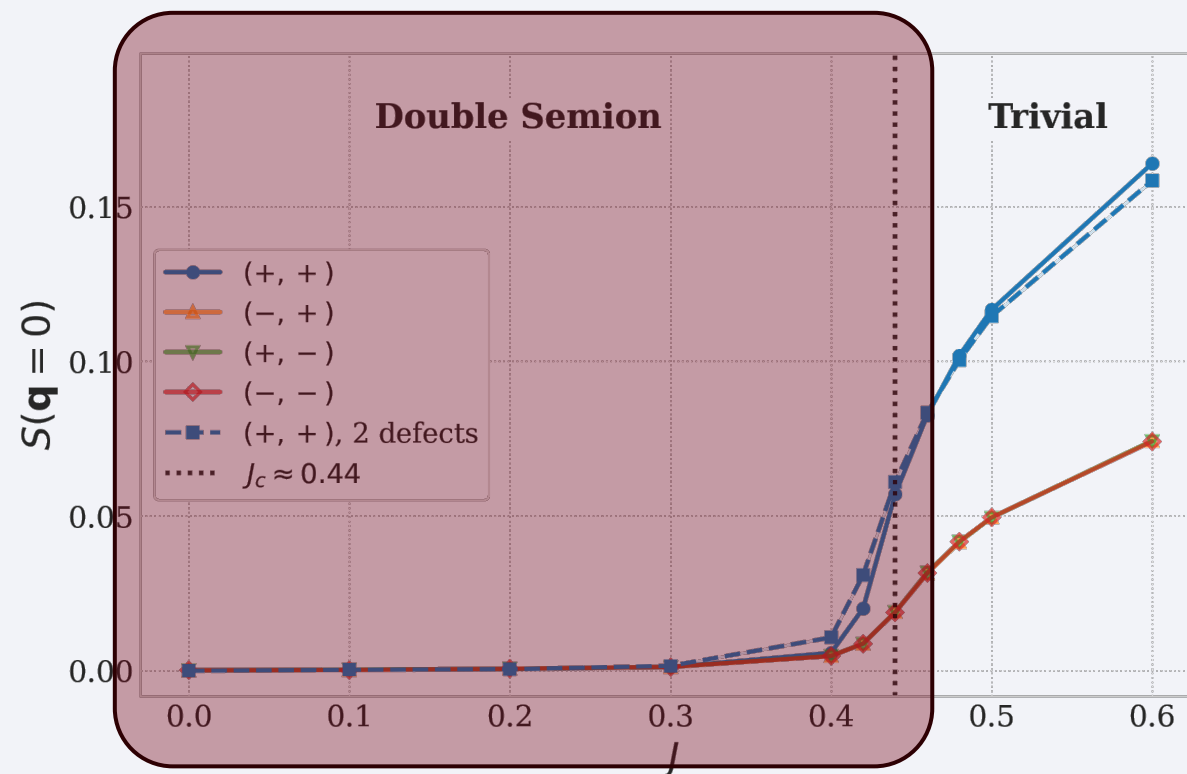


No-go theorems for topological phases



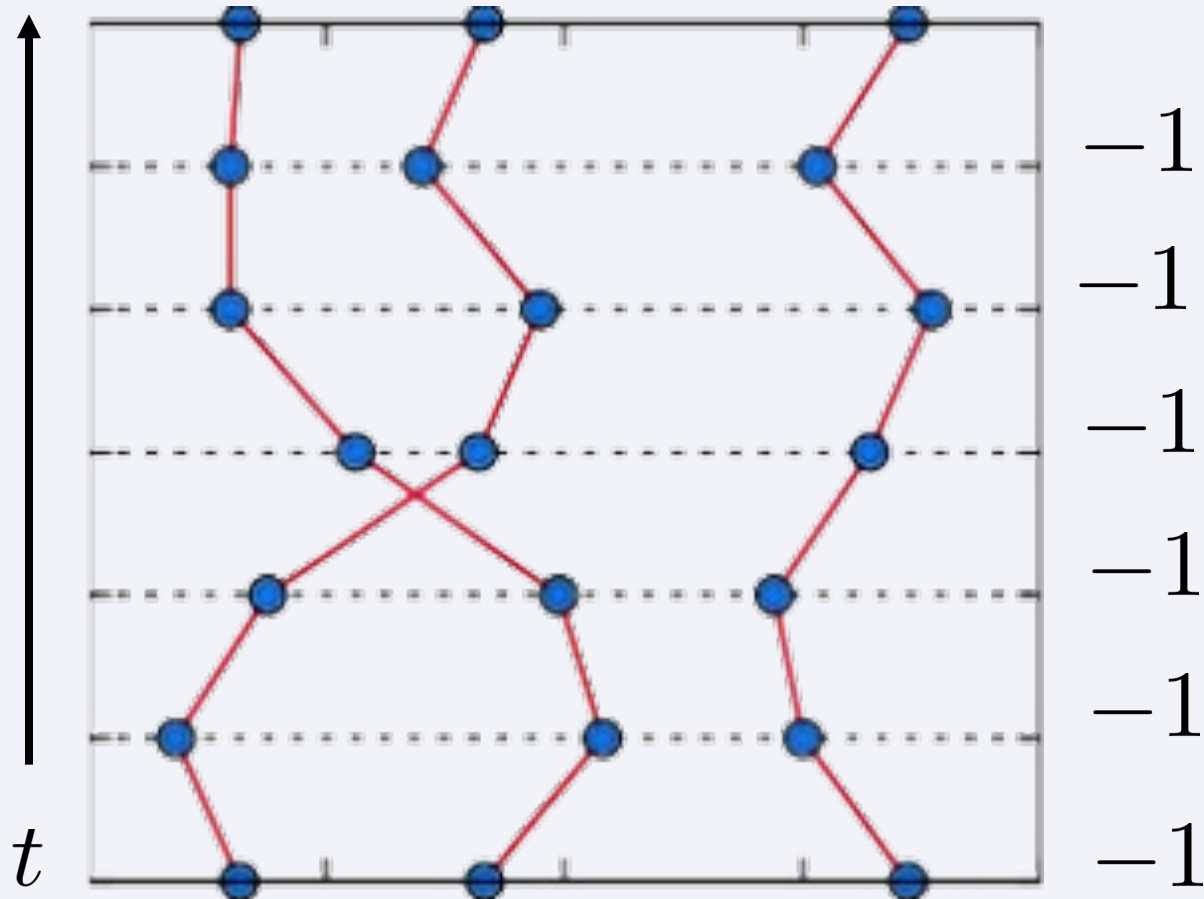
- Double semion topological order (Hastings 2016)
- **Most** topological orders (Ringel 2017, Seo 2025)
- “*We analyze the Gauss sums of 405 different topological orders classified up to rank 12 [...] and find that 398 suffer from the intrinsic sign problem.*”

A counterexample?



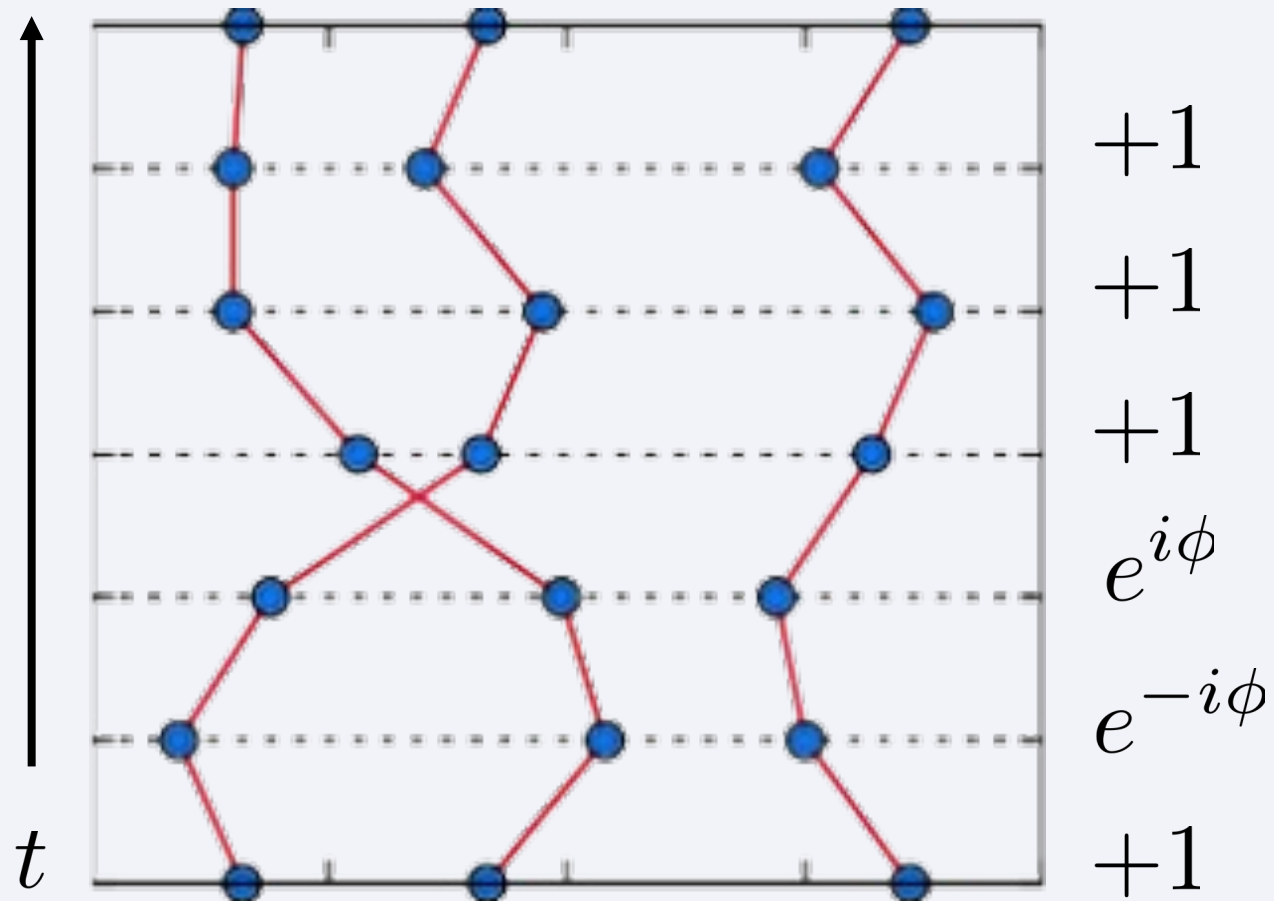
Simulated with **completely conventional** methods

Key mechanism: global cancellation of signs

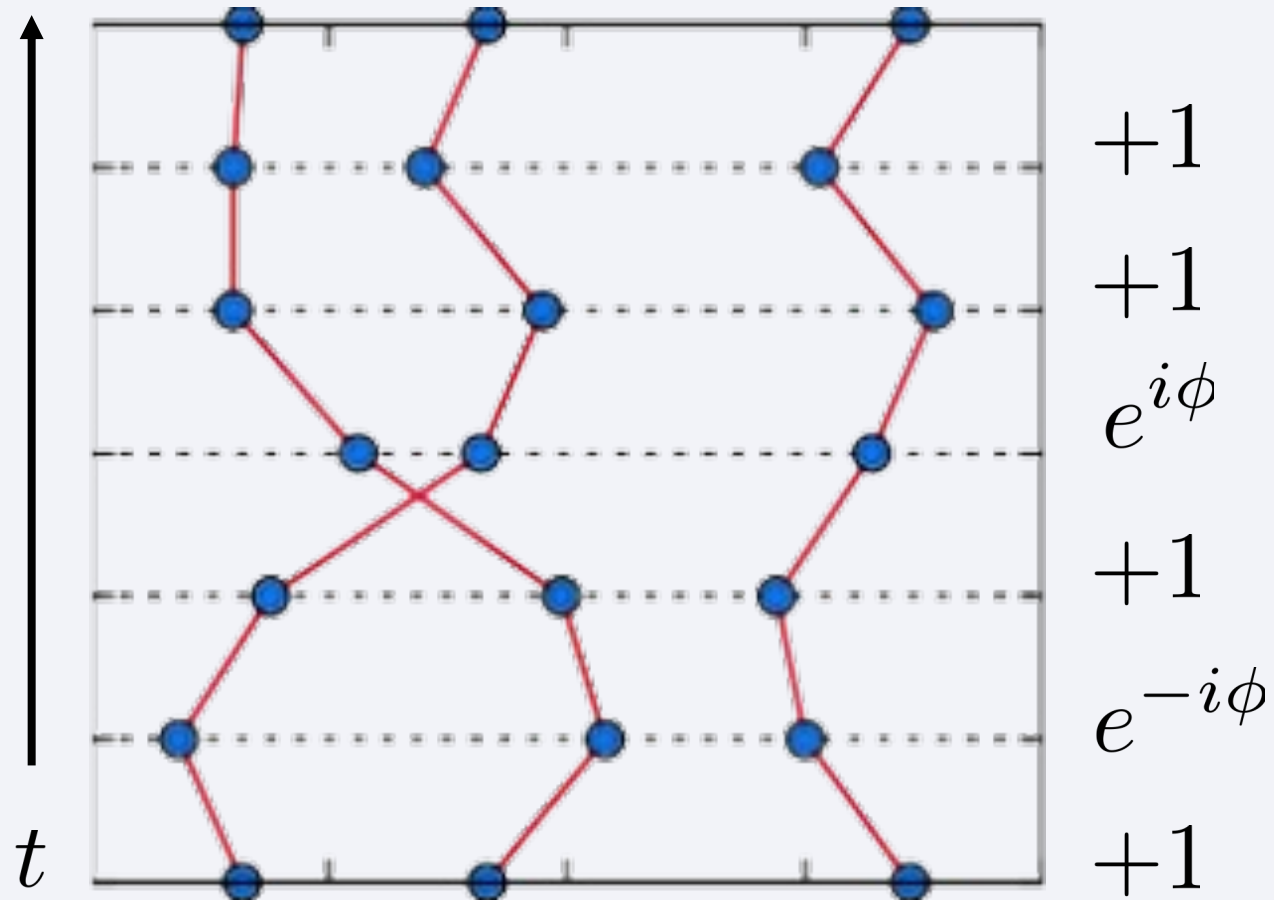


Sign problem-free requires **global** cancellations

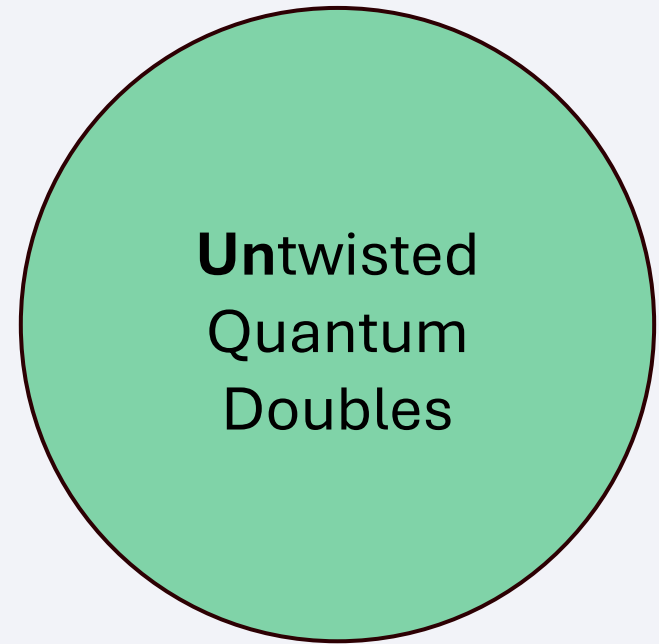
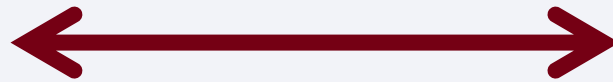
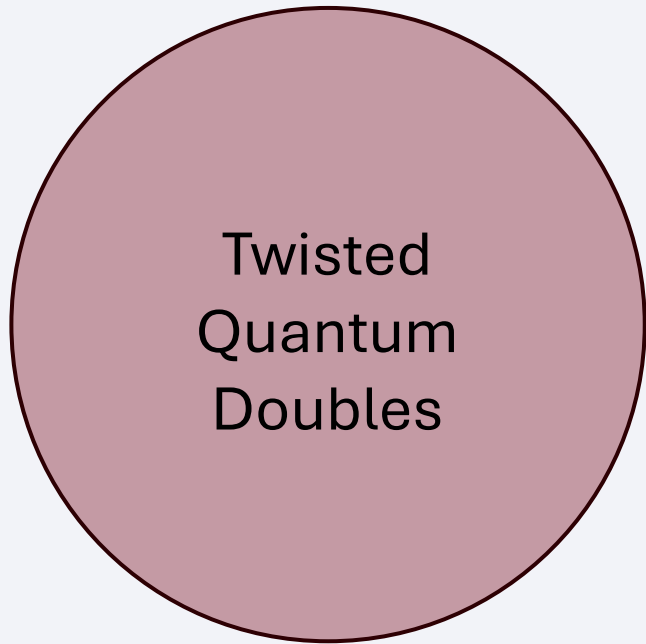
General theory: sign problem-preserving transformations



General theory: sign problem-preserving transformations



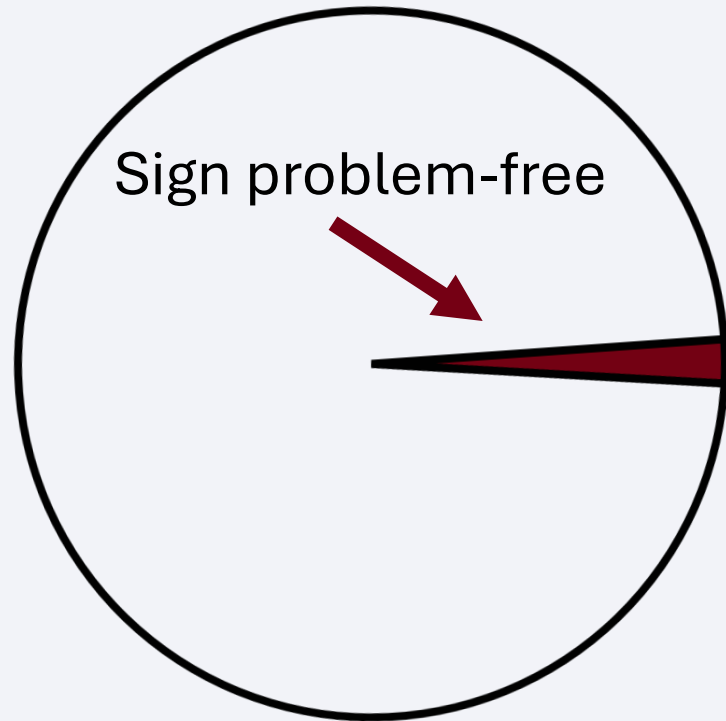
General theory: sign problem-preserving transformations



Previously thought to be
fundamentally inaccessible

What are the true limits to simulatability?

Before:

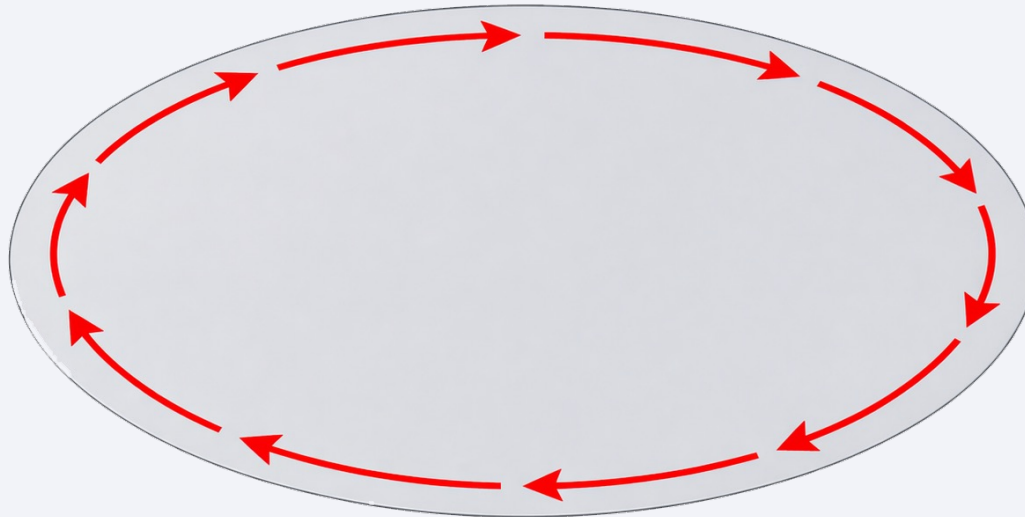


Now:



What are the *true* limits to simulatability?

- Refined criteria: sign problem-free models are *stoquasticizable*
- Are there obstacles to stoquasticizability?
- *Chiral* phases are an obstruction!



Thank you, Neil and Jane



A young man with short brown hair, wearing a black t-shirt with a 'Jack Flores' graphic and blue jeans, stands in front of a large projection screen. He is holding a small object in his hands. The screen displays a presentation slide with a light blue background. The slide has a purple header bar with the title 'Rutherford Scattering Detection through Gold Foil'. Below the title, the text 'Henry Shackleton' and 'May 15, 2017' are displayed. At the bottom of the slide, there is a purple bar with the text 'Henry Shackleton', 'Rutherford Scattering', 'May 15, 2017', and '1 / 19'. The video player interface at the bottom shows the MIT logo, the title 'Experimental Physics I: Final Presentation: Rutherford Scattering Detection through Gold Foil', the MIT OpenCourseWare logo with '6.29M subscribers', a 'Subscribe' button, and engagement icons for likes (304), dislikes, share, save, and a menu icon.

Rutherford Scattering Detection through Gold Foil

Henry Shackleton

May 15, 2017

Henry Shackleton Rutherford Scattering May 15, 2017 1 / 19

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Thank you, Neil and Jane



Twisted Quantum Doubles are Sign Problem-Free

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Poster #32

Many phases of matter
thought to be *fundamentally*
inaccessible to quantum
Monte Carlo methods are
actually simulatable!

